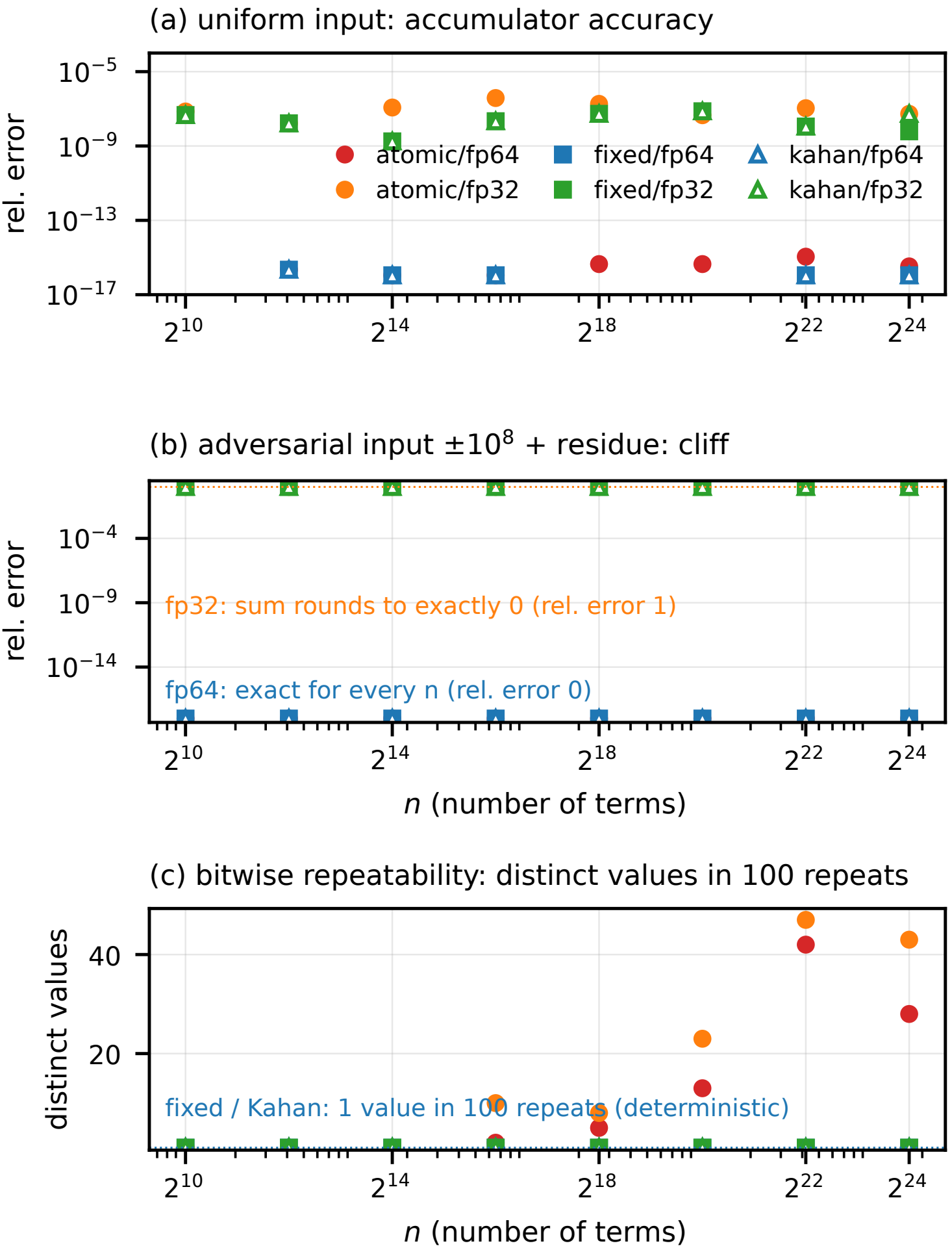
(a) uniform input: accumulator accuracy
rel. error
10^{-5}
10^{-9}
10^{-13}
10^{-17}
atomic/fp64
atomic/fp32
fixed/fp64
fixed/fp32
kahan/fp64
kahan/fp32
2^{10}
2^{14}
2^{18}
2^{22}
2^{24}
(b) adversarial input $\pm 10^8$ + residue: cliff
rel. error
10^{-4}
10^{-9}
10^{-14}
fp32: sum rounds to exactly 0 (rel. error 1)
fp64: exact for every n (rel. error 0)
2^{10}
2^{14}
2^{18}
2^{22}
2^{24}
n (number of terms)
(c) bitwise repeatability: distinct values in 100 repeats
distinct values
40
20
fixed / Kahan: 1 value in 100 repeats (deterministic)
2^{10}
2^{14}
2^{18}
2^{22}
2^{24}
n (number of terms)